Siddharth Saminathan

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siddharthsaminathan

Profile

AI Product Engineer with 4+ years building and shipping production AI systems that automate mission-critical enterprise workflows, from AI-driven quality management (DFMEA. PFMEA, Control Plans, 8D) to real-time AI products at scale. I take systems from concept to production and optimize them relentlessly: latency, cost, reliability, and user outcomes. Multi-agent orchestration, tool-driven architectures, and memory systems. What I build doesn't just respond, it improves over time.

Skills

Programming	Python · SQL
AI Systems	LLM APIs · RAG pipelines · embeddings · prompt engineering · tool calling · multi-model routing · agentic workflows · LangChain/LangGraph
Backend	FastAPI · Redis (caching, queues) · PostgreSQL · REST APIs · async pipelines · WebSockets. SSE streaming
Systems Design	Stateful systems · workflow orchestration · memory systems · context modeling · distributed pipelines
Infra	Docker · AWS · Fly.io · logging · tracing · metrics · performance optimization
Data	ETL pipelines · data validation · Power BI · Tableau

Work Experience

AI Engineer

Omnex Systems, Chennai
Jul 2025 - Present

Omnex is an **Enterprise AI Quality Platform (EQMS)** serving global manufacturing. I built their agentic AI system for end-to-end manufacturing quality digitalization, automating the core quality engineering workflows that automotive, aerospace, and industrial OEMs run on.

- Built a production multi-agent AI system that automates end-to-end manufacturing quality workflows: DFMEA (Design Failure Mode), PFMEA (Process Failure Mode), Control Plans, and 8D corrective action, reducing manual engineering effort and improving consistency across global supply chains.
- Designed a tool-driven orchestration layer that dynamically selects tools, routes data across internal and external enterprise systems, and executes multi-step quality workflows (generate to validate to update to act) with full traceability.
- Enabled context-aware quality decisioning: the system maps inputs to production hierarchies (part/process/plant), retrieves relevant historical quality cases, and suggests structured corrective actions and process updates.
- Improved system performance and reliability, reduced workflow latency (~12.5s to ~8s), built validation layers and retry logic, and instrumented full observability (logging, tracing, metrics) for production manufacturing environments.

Co-Founder / AI Engineer

Emote (emotenow.app)
Apr 2024 - Present

- Built and iterated a production AI system that learns from user interactions over time, modeling context, relationships, and behavioral patterns across sessions to improve response quality and continuity
- Designed and scaled the end-to-end backend architecture (FastAPI, Redis, PostgreSQL, real-time pipelines, multi-model routing), enabling low-latency, stateful AI interactions in production
- Drove core system performance and efficiency, reducing latency (~15s to ~3s) and Inference cost (~$1.20 to ~$0.023) through pipeline optimization, caching, and model selection
- Owned the full product loop (0 to 1 to iteration): scaled to 450+ users with 40+ power users, improved activation (20% to 75%) and retention (5% to 13-15%) by continuously shipping improvements based on real user behavior and feedback.

Data Engineer

Akkodis AB, Sweden
Nov 2022 - Nov 2024

- Led migration from Azure to PostgreSQL, automating ETL processes using Docker, Argo, and Python.
- Developed and optimized Cron jobs to enhance workflow efficiency.
- Improved Power BI dashboards and metrics pipelines, resulting in 5% increase in business reporting accuracy.
- Constructed automated validation layers to ensure data quality and integrity across enterprise reporting systems.
- Enhanced ETL workflows through Docker containerization and PostgreSQL optimization.

Data Analyst

Positive Integers Pvt Ltd, Chennai
Sep 2019 - May 2020

- Analyzed financial datasets related to loan portfolios, credit risk, and customer behavior patterns for an NBFC
- Developed interactive dashboards using Tableau and Grafana to visualize business KPIs and financial metrics
- Constructed SQL-based ETL workflows.

Key Achievements

- **Built Emote from zero:** Took product from vision to production, grew to 450+ users, achieved 13-15% weekly retention through continuous iteration.
- **Latency reduction:** Drove inference latency from ~15s to ~3s and cost from ~$1.20 to ~$0.023 per session at Emote.
- **Enterprise migration:** Led Azure to PostgreSQL data platform migration at Akkodis for Ericsson systems; recognized for improving reporting reliability and delivery.
- **Agentic manufacturing quality system:** Built production multi-agent AI for enterprise quality management (EQMS) at Omnex Systems, automates DFMEA, PFMEA, Control Plans, and 8D across global manufacturing supply chains (12.5s to 8s latency).

Education

- **M.Sc. Statistics and Machine Learning** - Linköping University, Sweden (2020-2022)
- **B.Tech Computer Science and Engineering** - SRM University, India (2016-2020)

Master's Thesis

Developed a machine learning pipeline utilizing autoencoders and classifiers to predict colorectal cancer metastasis from RNA sequencing data. Employed transfer learning, PCA, and network analysis for biomarker detection in high-dimensional biological datasets.

Languages

English (Fluent) · Tamil · Hindi